LEARNING HIERARCHICAL AND GEOMETRY-AWARE GRAPH REPRESENTATIONS FOR TEXT-TO-CAD

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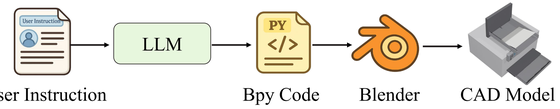
TEXT-TO-CAD & CHALLENGES

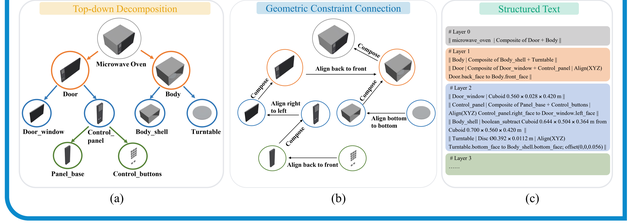
Text-to-CAD aims to translate natural-language instructions into executable CAD programs that produce editable 3D models through a valid sequence of CAD operations.

- **Long-horizon generation:** Text-to-CAD requires generating a long sequence of interdependent operations, where small errors in early steps can propagate through later stages and invalidate the final assembly.
- **Flat text-to-code decoding:** Existing methods often map text directly into code without explicit assembly hierarchy or geometric relations, making complex multi-part CAD generation prone to assembly failures.



MOTIVATION

- **What is needed is an intermediate structure:** It should explicitly capture both the assembly hierarchy of parts and their geometric relations.
- **Graph as a structural prior:** It encodes multi-level parts as nodes and geometric relations as edges, helping narrow the search space for long-horizon CAD generation.



GRAPH-CAD FRAMEWORK

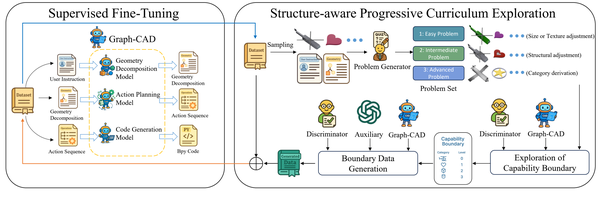
Graph-CAD transforms a user instruction into executable CAD code through a three-stage pipeline.

- **Stage 1: Geometry Decomposition.** The instruction is first converted into a geometric decomposition graph that explicitly represents part hierarchy and geometric constraints.
- **Stage 2: Action Planning.** Based on the predicted graph, the model determines an ordered sequence of CAD operations.
- **Stage 3: Code Generation.** The operations are finally translated into executable code for CAD model generation.

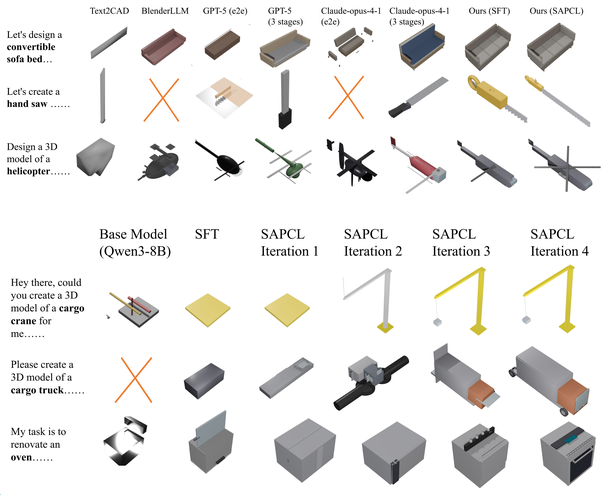
SAPCL

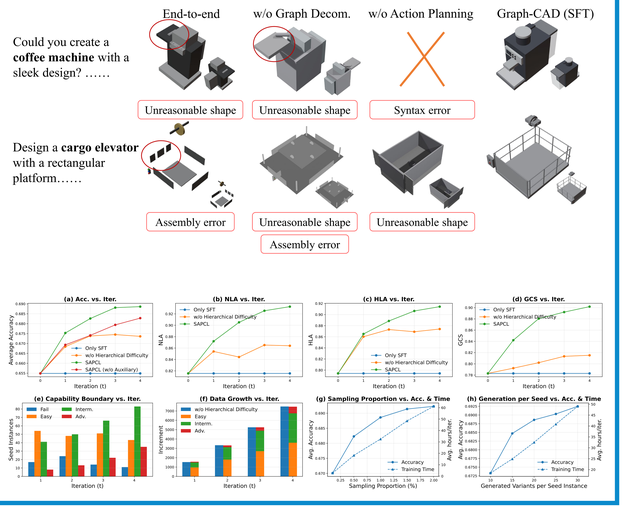
SAPCL improves Graph-CAD by progressively expanding its capability boundary on complex CAD assemblies.

- **Difficulty-aware curriculum:** Starting from seed examples, SAPCL constructs graded variants with easy, intermediate, and advanced difficulty levels.
- **Capability boundary exploration:** The model is evaluated on these variants to identify the highest difficulty level it can reliably handle.
- **Boundary-focused expansion:** New training samples are synthesized and validated near this boundary, enabling the model to progressively master more complex and highly constrained CAD structures.



QUALITATIVE COMPARISONS





QUANTITATIVE COMPARISONS

Models	CADBench-Sim							CADBench-Wild						
	Attr.↑	Spat.↑	Inst.↑	Avg.↑	E_{syntax}↓	CLIP↑	GCS↑	Attr.↑	Spat.↑	Inst.↑	Avg.↑	E_{syntax}↓	CLIP↑	GCS↑
Specifically Text-to-CAD open-source models														
BlenderLLM	0.6893	0.6953	0.3650	0.5832	2.4%	0.6409	0.5513	0.6782	0.6363	0.4581	0.5909	5.3%	0.6056	0.4983
Text2CAD	0.3278	0.2084	0.0446	0.1936	6.6%	0.5707	-	0.4198	0.3082	0.1323	0.2868	14.0%	0.5211	-
CADFusion	0.3566	0.2258	0.0674	0.2166	6.2%	0.5578	-	0.3822	0.3716	0.1496	0.3011	11.5%	0.5278	-
General-purpose Large Language Models														
Qwen-Plus	0.3604	0.3777	0.2072	0.3151	48.4%	0.3362	0.2379	0.2596	0.2722	0.1951	0.2423	61.0%	0.2446	0.1305
Llama-3.1-405b	0.3302	0.3355	0.1537	0.2731	36.4%	0.3943	0.3269	0.3331	0.3530	0.1943	0.2934	47.2%	0.3242	0.2903
Deepseek-r1	0.4124	0.4366	0.2179	0.3556	19.2%	0.5011	0.5556	0.4814	0.5141	0.3735	0.4564	20.50%	0.4858	0.4275
Gemini-2.5-pro	0.2173	0.2180	0.1565	0.1972	42.4%	0.2050	0.4048	0.2002	0.1880	0.1667	0.1850	48.7%	0.1750	0.2584
GPT-5	0.7013	0.7347	0.4250	0.6203	2.8%	0.6449	0.3846	0.6858	0.7091	0.5595	0.6515	5.5%	0.6003	0.4017
Claude-opus-4-1	0.7216	0.7368	0.5403	0.6662	7.4%	0.6151	0.4932	0.6847	0.7218	0.5997	0.6687	14.5%	0.5550	0.5062
Graph-CAD (Ours)														
Graph-CAD (SFT)	0.7295	0.7265	0.4733	0.6431	2.4%	0.6544	0.7830	0.6944	0.7270	0.5861	0.6692	4.5%	0.6358	0.8025
Graph-CAD (SAPCL)	**0.7681**	**0.7423**	**0.5546**	**0.6883**	**2.0%**	**0.6693**	**0.9018**	**0.7695**	**0.7590**	**0.6057**	**0.7114**	**2.5%**	**0.6577**	**0.8943**
SAPCL vs SFT	(5.29%↑)	(2.17%↑)	(17.18%↑)	(7.03%↑)		(2.28%↑)	(15.17%↑)	(10.82%↑)	(4.40%↑)	(3.34%↑)	(6.31%↑)		(3.44%↑)	(11.44%↑)